

Model Risk Validation Report

Full Technical & Quantitative Analysis

GENERATED AT
2026-07-05

MONITOR ID
mon_c1027f347599

RUN ID
run_ccc25501...

OVERALL
HEALTH
22.5/100

CRITICAL

Key Performance Metrics

Metric	Category	Value	Threshold	Status
Brier Skill Score	calibration	-0.0609	N/A	OK
Calibration Ratio	calibration	0.5935	N/A	CRITICAL
Row Count	data_quality	197.0000	N/A	OK
Duplicate Row Count	data_quality	0.0000	N/A	OK
Null Rates	data_quality	N/A	N/A	OK
Invalid Probability Count	data_quality	0.0000	N/A	OK
Invalid Target Count	data_quality	0.0000	N/A	OK
Monitoring Readiness Summary	data_quality	N/A	N/A	OK
Score-PD Coherence	data_quality	-0.0422	N/A	CRITICAL
Band Consistency	data_quality	0.8680	N/A	WARNING
DPD Buckets	delinquency	N/A	N/A	OK
Score PSI	drift	N/A	N/A	SKIPPED
ROC AUC	performance	0.3341	N/A	CRITICAL

METRIC	CATEGORY	VALUE	THRESHOLD	STATUS
Gini Coefficient	performance	-0.3319	N/A	CRITICAL
KS Statistic	performance	0.3563	N/A	OK
Decile Table	performance	N/A	N/A	OK
Score Distribution Analysis	performance	N/A	N/A	CRITICAL
Approval Rate	strategy	0.7716	N/A	OK
Overall Bad Rate	strategy	0.0558	N/A	OK

Statistical Tests

TEST NAME	STATISTIC	P-VALUE	CONCLUSION	RESULT
Hosmer-Lemeshow Test	5.6961	0.6607	Fail to reject H_0 — distributions are similar	
Binomial Test: Band 600-700	5.0000	0.6607	Fail to reject H_0 — distributions are similar	
Binomial Test: Band 700-800	4.0000	0.6607	Fail to reject H_0 — distributions are similar	
Binomial Test: Band <600	1.0000	0.9810	Fail to reject H_0 — distributions are similar	
Fisher Exact Test: Band 800+	1.4091	0.6942	Fail to reject H_0 — distributions are similar	
Mann-Whitney U: Approved vs Declined Scores	3354.0000	0.6942	Fail to reject H_0 — distributions are similar	

Fairness Analysis (Disparate Impact)

Summary: No direct adverse impact detected. Borderline: credit_car (DI=0.8308)

PROTECTED GROUP	VALUE	N	APPROVAL RATE	DI RATIO	STATUS
loan_purpose	credit_car	39	64.1%	0.83	WARN
income_band	75-100K	43	65.1%	0.84	WARN
home_ownership	OTHER	14	71.4%	0.93	OK
loan_purpose	debt_cons	36	75.0%	0.97	OK
income_band	<50K	66	75.8%	0.98	OK
home_ownership	MORTGAG	81	76.5%	0.99	OK
home_ownership	RENT	87	78.2%	1.01	OK
home_ownership	OWN	15	80.0%	1.04	OK
loan_purpose	other	41	80.5%	1.04	OK
loan_purpose	home_imp	37	81.1%	1.05	OK
income_band	100K+	31	83.9%	1.09	OK
loan_purpose	major_pur	44	84.1%	1.09	OK
income_band	50-75K	57	84.2%	1.09	OK

Vintage Analysis (Roll Rate)

Note: True roll rate matrix requires two time-snapshot datasets. This shows current DPD bucket distribution.

Severe Delinquency Rate: 23.86%

BUCKET	COUNT	PERCENTAGE	CUMULATIVE %
Current (0 DPD)	100	50.8%	50.8%
1-29 DPD	0	0.0%	50.8%
30-59 DPD	24	12.2%	62.9%
60-89 DPD	26	13.2%	76.1%
90+ DPD	47	23.9%	100.0%

Override Analysis

"Overridden accounts in this dataset show LOWER bad rate than model-approved accounts (0.0% vs 6.3%), suggesting underwriters are exercising sound judgment for this risk segment. Review override criteria."

MODEL DECLINES

35

OVERRIDES

26

OVERRIDE RATE

74.3%

OVERRIDE BAD RATE

0.0%

Conclusion: Overrides are performing better or equal to model approvals.

"Overridden accounts in this dataset show LOWER bad rate than model-approved accounts (3.3% vs 6.5%), suggesting underwriters are exercising sound judgment for this risk segment. Review override criteria."

MODEL DECLINES

73

OVERRIDES

60

OVERRIDE RATE

82.2%

OVERRIDE BAD RATE

3.3%

Conclusion: Overrides are performing better or equal to model approvals.

"Overridden accounts in this dataset show LOWER bad rate than model-approved accounts (3.6% vs 7.4%), suggesting underwriters are exercising sound judgment for this risk segment. Review override criteria."

MODEL DECLINES

OVERRIDES

107

84

OVERRIDE RATE

78.5%

OVERRIDE BAD RATE

3.6%

Conclusion: Overrides are performing better or equal to model approvals.

"Overridden accounts in this dataset show LOWER bad rate than model-approved accounts (3.8% vs 8.5%), suggesting underwriters are exercising sound judgment for this risk segment. Review override criteria."

MODEL DECLINES

136

OVERRIDES

105

OVERRIDE RATE

77.2%

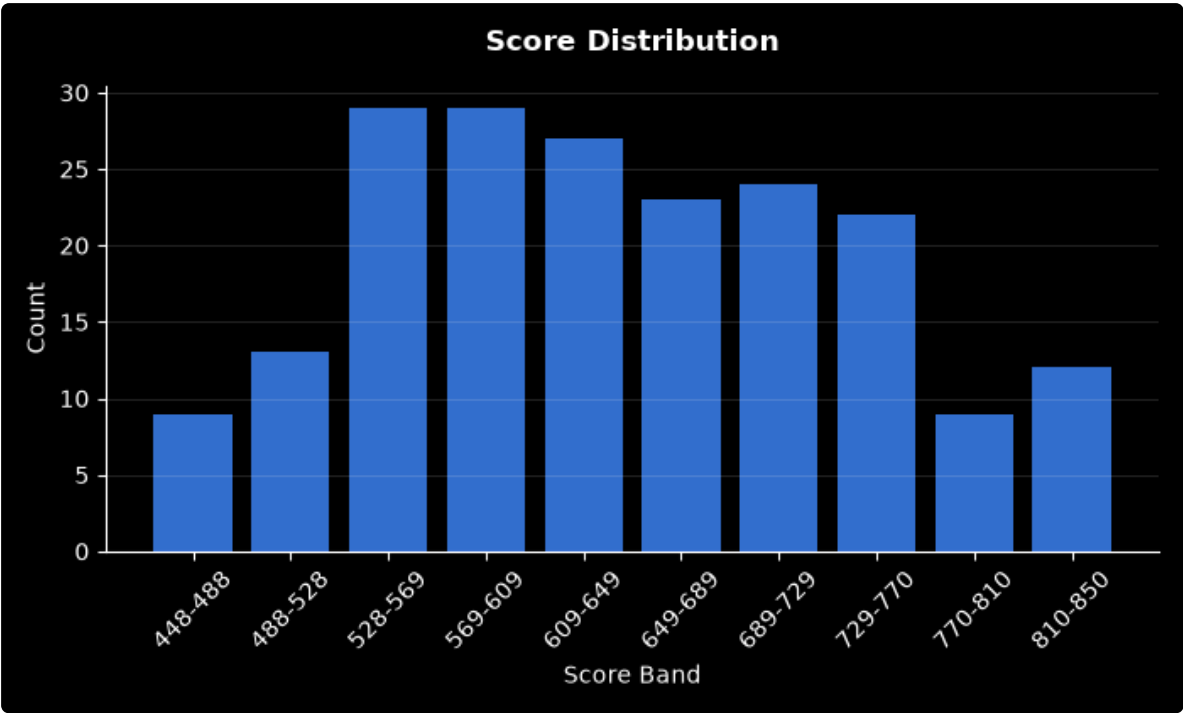
OVERRIDE BAD RATE

3.8%

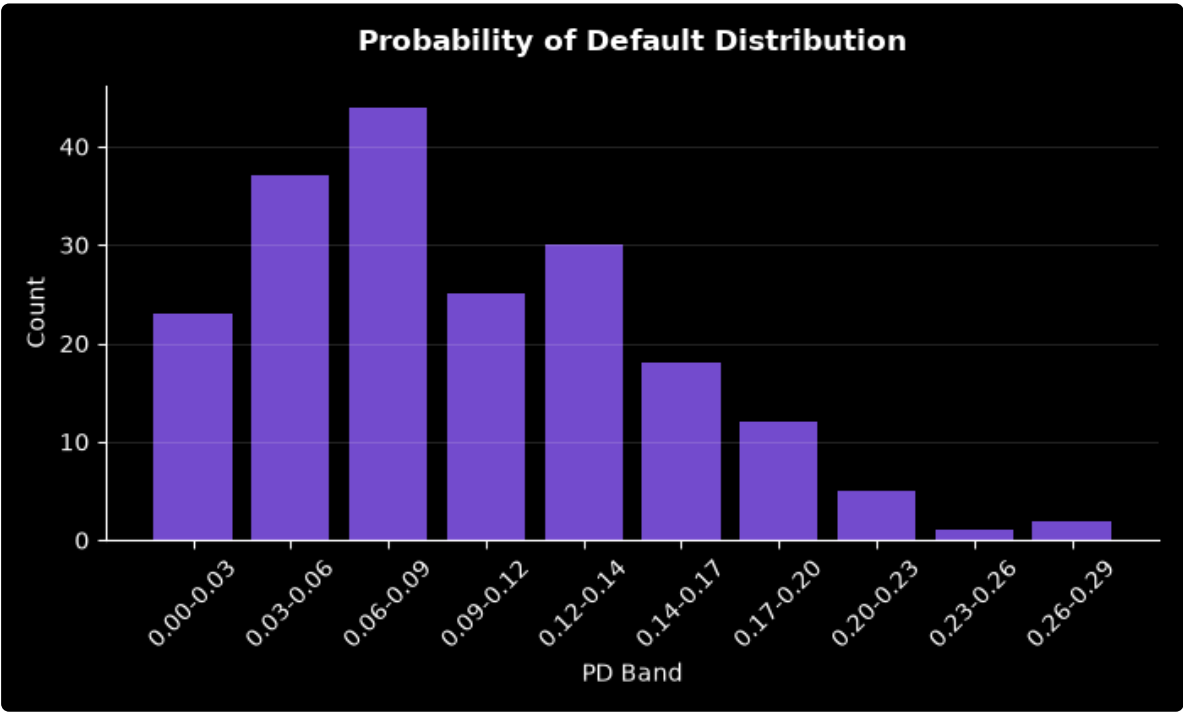
Conclusion: Overrides are performing better or equal to model approvals.

Model Visualizations

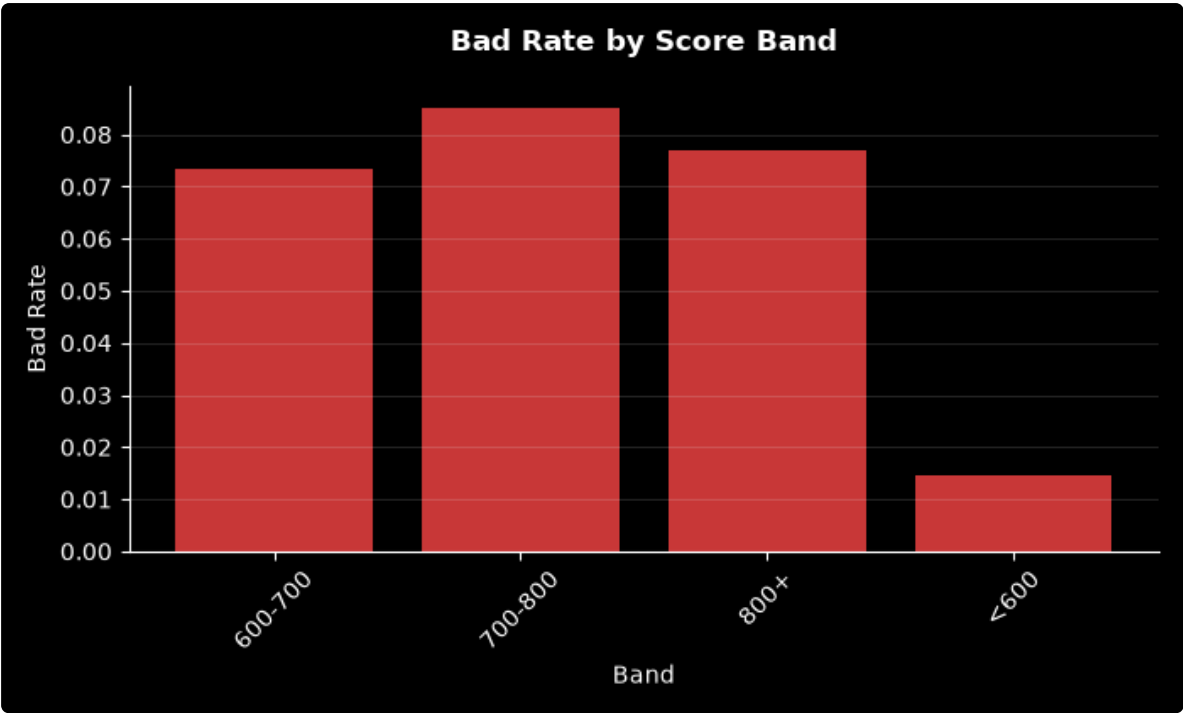
CHART_SCORE_DISTRIBUTION



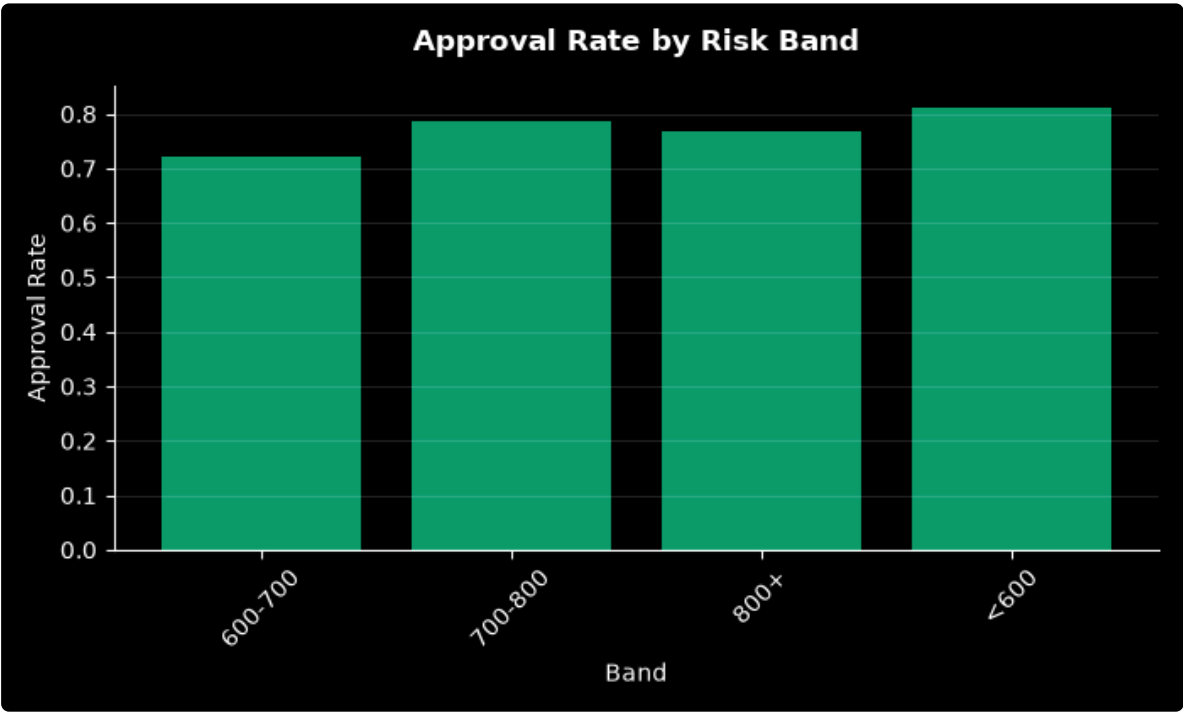
CHART_PD_DISTRIBUTION



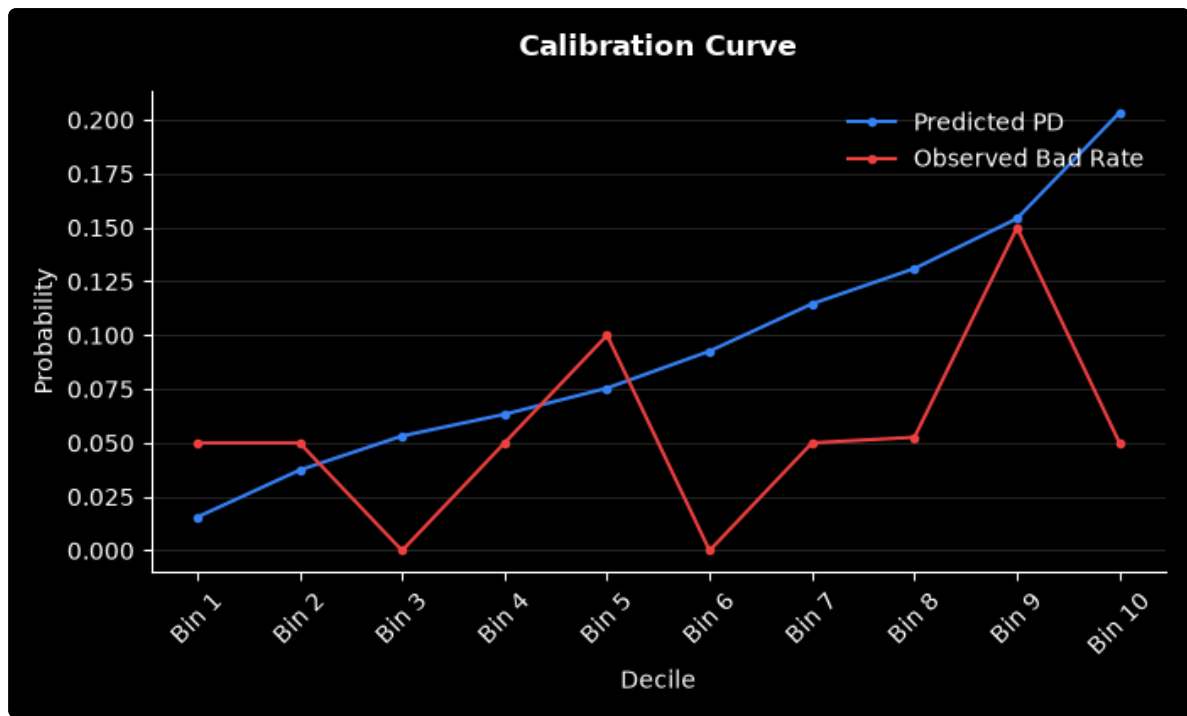
CHART_BAD_RATE_BY_SCORE_BAND



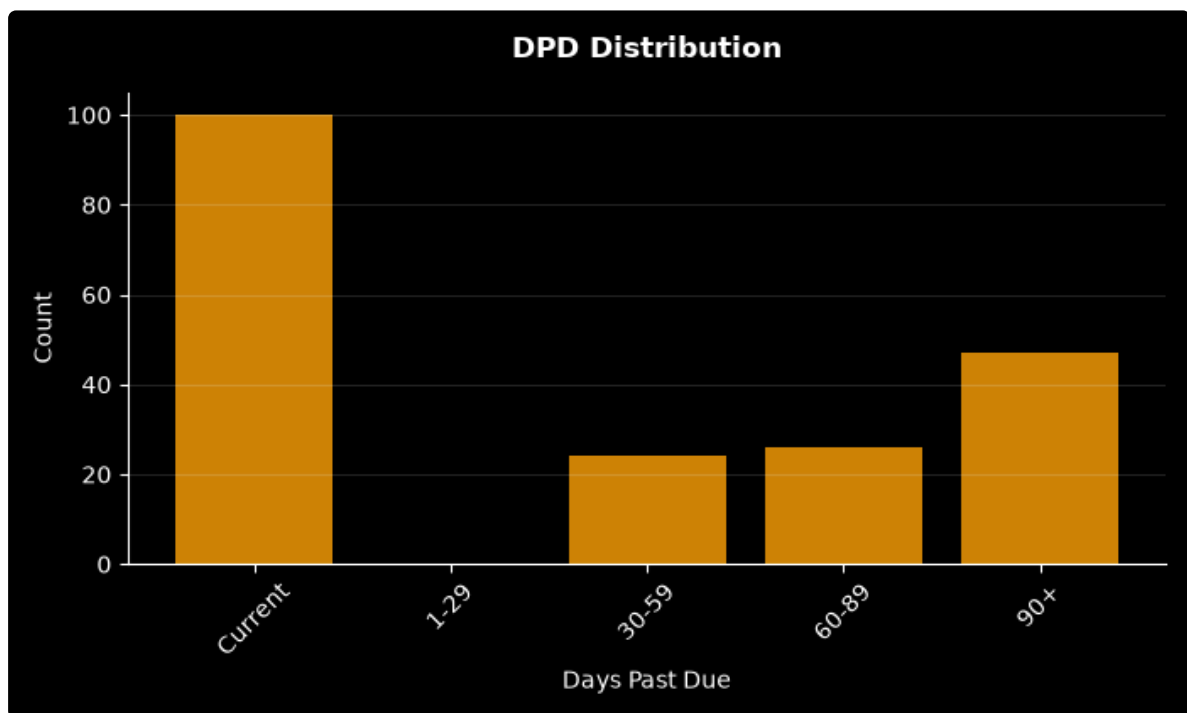
CHART_APPROVAL_BY_RISK_BAND



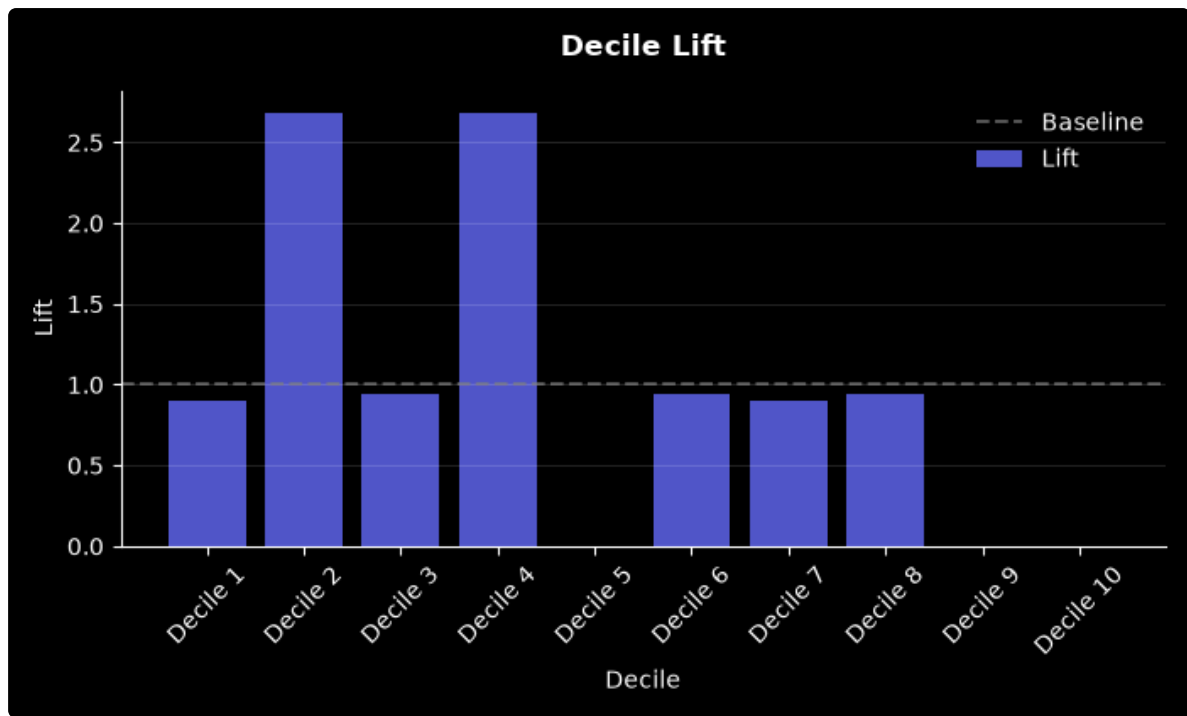
CHART_CALIBRATION



CHART_DPD_DISTRIBUTION



CHART_DECILE_LIFT



CHART_ROC_CURVE

